Customer Data Insights Report

Findings from the Excel to Insights Engine · customer_data.xlsx · generated 27 May 2025

Problem: a business has a spreadsheet of customer records and needs quick answers without manual analysis: how much do customers spend, how are they distributed, what drives their value, and where is the data unreliable? **Approach:** the Python tool loaded the Excel file and automatically ran summary statistics, correlation analysis, outlier detection, categorical analysis and charting. This report presents what it found.

Note on the data: *the analysed file is a synthetic sample dataset generated for testing the tool, so the numbers below demonstrate the method and are not real business results.*

200	15	478.92	0.81
customer records	columns analysed	mean total spend	frequency vs. value correlation

1. Key Findings

- **Spending is right-skewed.** Mean total spend is 478.92 against a median of 342.17, and the top spender reached 2,628.04. A small group of big spenders pulls the average up; 12 customers (6.42%) are statistical outliers.
- **Order value is steady.** Average order value has a mean of 95.99 and a standard deviation of 29.15, with only 2 outliers (1.00%).
- **Value follows purchase frequency.** Purchase frequency and customer value are strongly correlated (0.81), so increasing how often customers buy is the clearest lever in this data.
- **Segments are balanced.** Standard 37.0%, Premium 32.5% and Basic 30.5% of customers; no single segment dominates.
- **Social media is the top acquisition channel** at 24.5%, ahead of Email (19.0%), Direct (16.5%), Referral (14.0%) and Organic Search (13.5%).
- **Some data is missing.** Age is missing for 8 customers (4.00%), Total Spend for 13 (6.50%) and Days Since Last Purchase for 14 (7.00%).

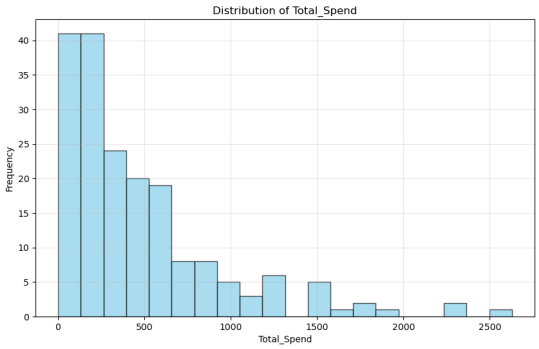
2. Dataset Overview

Item	Detail
File	customer_data.xlsx
Sheets	Sheet1
Rows	200
Columns	15
Numeric columns analysed	Customer_ID, Age, Total_Purchases, Total_Spend, Average_Order_Value, Days_Since_Last_Purchase, Customer_Tenure_Days, Purchase_Frequency, Customer_Value
Categorical columns	Gender, Location, Segment, Acquisition_Channel
Date column	Registration_Date (2020-05-28 to 2025-05-27)

3. Summary Statistics

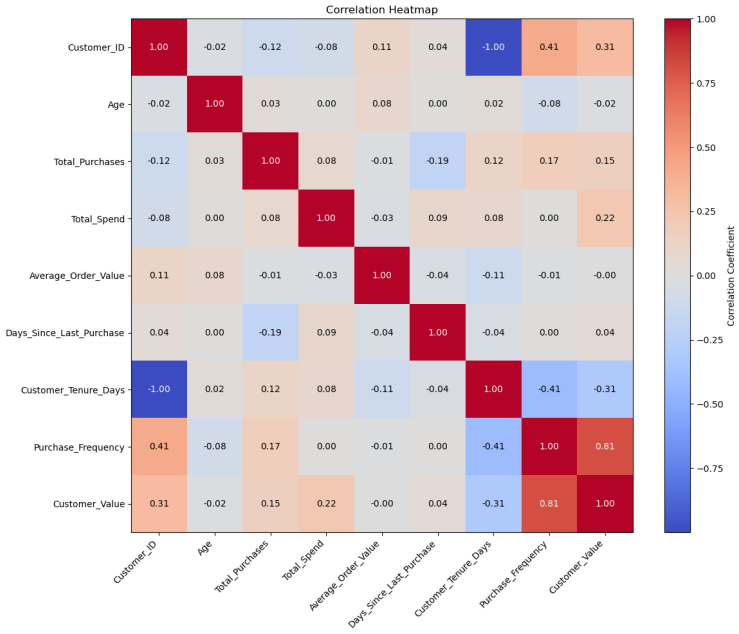
Column	Count	Mean	Std Dev	Min	25%	Median	75%	Max
Age	192	49.39	18.16	18.00	34.00	51.00	67.00	79.00
Total_Purchases	200	24.97	14.99	0.00	12.00	25.00	38.00	49.00
Total_Spend	187	478.92	473.47	1.54	159.41	342.17	621.46	2,628.04
Average_Order_Value	200	95.99	29.15	7.21	74.41	95.54	115.82	189.96
Days_Since_Last_Purchase	186	182.27	109.40	2.00	102.50	177.00	289.00	364.00
Customer_Tenure_Days	200	912.00	530.79	0.00	455.75	912.00	1,368.25	1,825.00
Purchase_Frequency	200	n/a*	n/a*	0.00	0.40	0.81	1.58	n/a*
Customer_Value	187	n/a*	n/a*	0.00	5.95	22.30	61.98	n/a*

** Purchase_Frequency and Customer_Value contain infinite values, caused by customers with 0 tenure days (division by zero). The mean, standard deviation and maximum are therefore undefined; the quartiles are unaffected. Customer_ID is an identifier and is left out of this table.*



4. Correlation Analysis

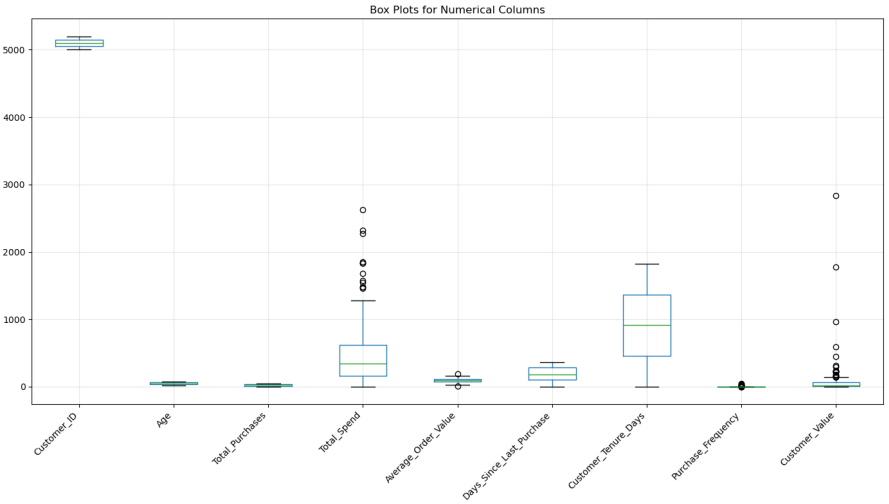
The tool flagged two pairs above its 0.7 threshold. **Purchase_Frequency and Customer_Value (0.81)** move together, which is expected because customer value is calculated from purchase frequency. **Customer_ID and Customer_Tenure_Days (-1.00)** is an artifact of how the sample data was built (IDs and tenure were generated in opposite order), not a real relationship.



5. Outlier Detection

Outliers were found with the IQR method (values beyond 1.5 times the interquartile range).

Column	Outliers	Share of data	Sample values
Total_Spend	12	6.42%	1,481.85 · 1,835.93 · 1,458.70 · 2,272.61
Average_Order_Value	2	1.00%	7.21 · 189.96
Purchase_Frequency	22	11.00%	3.49 · 3.43 · 4.04 · 5.90
Customer_Value	22	11.76%	169.15 · 226.74 · 299.60 · 313.17



6. Customer Profile

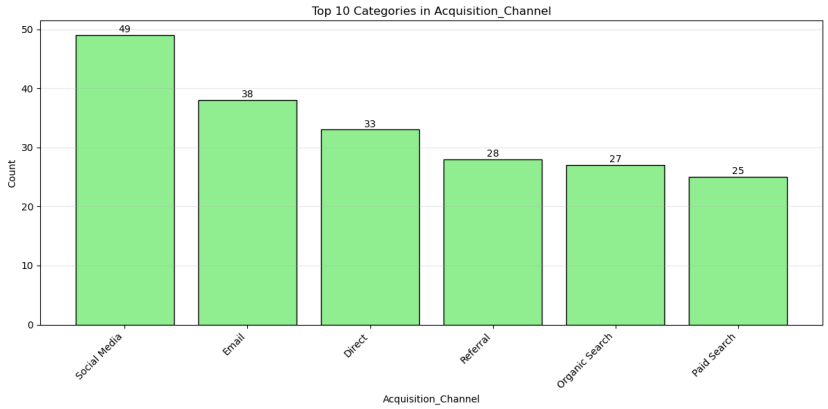
Categorical columns were summarised by their most common values.

Segment	Customers	Share
Standard	74	37.0%
Premium	65	32.5%
Basic	61	30.5%

Gender	Customers	Share
Female	52	26.0%
Male	51	25.5%
Prefer not to say	50	25.0%
Non-binary	47	23.5%

Acquisition channel	Customers	Share
Social Media	49	24.5%
Email	38	19.0%
Direct	33	16.5%
Referral	28	14.0%
Organic Search	27	13.5%

Customers are spread across 30 locations, with no city above 5% (the largest, City-26, has 10 customers). Gender has four roughly equal groups.



7. Registrations Over Time

Customers registered between 28 May 2020 and 27 May 2025, spread evenly across weekdays and months. Each full year from 2021 to 2024 added 40 customers; 2020 and 2025 are partial years.

Year	2020	2021	2022	2023	2024	2025
New customers	24	40	40	40	40	16

8. Conclusions and Limitations

- **What the analysis shows:** in this dataset, how often customers purchase matters most for customer value, spending is concentrated in a few high spenders, and about 4–7% of key fields are missing and should be cleaned before any modelling.
- **Data limitations:** the data is synthetic and randomly generated, so segment, gender, channel and location splits are near-uniform by design; real data would show clearer differences.
- **Tool limitations found:** the date analysis in the raw tool output also treated numeric columns (such as Customer_ID and Age) as dates (all showing 1970-01-01). Those entries are meaningless and were left out of this report; only Registration_Date is a true date column.